Introduction Of Devops

What Exactly Is DevOps?

Let's understand DevOps from the beginning,

First, understand a simple problem.

Imagine you are a software developer. You create a website for a college.

You have completed:

- Login and registration
- Student dashboard
- Admin dashboard
- Database connectivity
- All the required features

Now, your website works perfectly on your laptop.

But your college wants to make this website available to 1,000 students through the internet.

Now what?

You need to:

1. Put your application on a server.
2. Configure the server.
3. Connect the database.
4. Make your application accessible through the internet.
5. Monitor whether the application is working.
6. Fix problems if the application crashes.
7. Update the application whenever you add new features.

These activities involve much more than writing code.

This is where DevOps comes into the picture.

What is DevOps?

DevOps = Development + Operations

Dev	+	Ops
Development Operations		
Writing and developing software		Deploying and maintaining software

= DevOps

A culture and set of practices that bring development and IT operations together.

DevOps helps developers and operations teams work together to deliver software faster, with fewer errors and continuous improvements.

What does a DevOps Engineer actually do?

Let's take the example of your college website.

A DevOps Engineer helps automate and manage the journey from code development to running software.

DevOps workflow

1. **Code**

   Developer writes code using Python, Java, etc.

   Example: VS Code, Git

2. **Build**

   Prepare the application so it can run.

Example: Maven, Gradle

3. **Test**

   Automatically check whether the application works correctly.

   Example: pytest, JUnit

4. **Deploy**

   Make the application available on a server.

   Example: Docker, Kubernetes, AWS

5. **Monitor**

   Check application performance, errors and availability.

The process continues whenever developers make changes to the application.

Important DevOps tools you should know

1. Git

Tracks changes in your code.

1. GitHub

Stores and shares your code online.

1. Linux

An operating system commonly used for servers.

1. Docker

Packages applications with their dependencies into containers

1. Jenkins

Automates building, testing and deployment.

1. Kubernetes

Manages and coordinates containers at scale.

1. AWS

Provides cloud services such as servers, storage and networking.

5. How DevOps works in a real company

Suppose a company has a shopping application.

Without automation:

- Developer writes code.
- Someone manually tests it.
- Someone manually uploads it to a server.
- Deployment may take hours.
- Errors can happen during deployment.

With DevOps:

- Developer pushes code to GitHub.
- Automated pipeline starts.
- Application gets built.
- Automated tests run.
- If tests pass, application gets deployed.
- Monitoring tools check application health.

This automated process is commonly called CI/CD (Continuous Integration and Continuous Delivery/Deployment).

Is DevOps only about tools?

No! This is very important.

DevOps includes three major things:

1. Knowledge: Linux, networking, cloud, scripting.
2. Tools: Git, Docker, Jenkins, AWS, Kubernetes.

3. Practices: Automation, collaboration, continuous testing, deployment and monitoring.

Learning tools without understanding the concepts will not make you a good DevOps Engineer.